

SurrealDB: Indexed ORDER BY leaks the value ordering of a SELECT-restricted field

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1 Total Advisories	0 Critical	0 High	1 Medium
MEDIUM Severity	N/A CVSS / Risk Score	TLP:CLEAR TLP Classification	PRO Customer Tier

Executive Summary

A field can be hidden from a user with a field-level SELECT permission (`DEFINE FIELD code ON secret PERMISSIONS FOR select WHERE owner = $auth.id`). When that field is indexed, a record user who cannot read it could still recover the relative ordering of its values across every record by issuing `ORDER BY <field>`: the field came back `null` as intended, but the rows were returned in the hidden values' true sorted order. To satisfy the sort, the planner selects the field's index and walks it in value order; the field-level permission is applied later, when the row is projected, so the value is nulled but the row order already encodes it. The guard that withholds restricted fields from the `WHERE` path was never applied to `ORDER BY`. ## Impact What an attacker **can** do: - As a record (scope) user with table SELECT, learn the relative ordering of a field hidden by a field-level SELECT permission, across other users' records, by ordering on it when an index covers the field — the

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High-priority advisories ranked by CVSS score, exploitability, and actor attribution.

MITRE ATT&CK® v15 Mapping

Technique density score: 0.0 · Techniques observed: 0 · Tactics covered: 0

MITRE ATT&CK techniques pending analyst enrichment. Deploy detection rules from Section 6 to identify related activity.

Attribution analysis across advisories in this report period.

Indicators of Compromise (IOCs)

IOC extraction for this advisory is pending enrichment pipeline completion. PRO subscribers receive real-time IOC push to SIEM/SOAR via the SENTINEL APEX webhook at: **GET /api/v1/intell/iocs?advisory={id}**. Full IOC packages include IPv4, domain, SHA256, and URL indicators with confidence scores, first-seen timestamps, and STIX 2.1 formatted bundles.

Detection Engineering

Deploy these production-ready detection rules into your SIEM platform to identify exploitation attempts in real time. Compatible with Microsoft Sentinel, Splunk ES, and any Sigma-compatible platform.

Sigma Rule - Webserver / Process Monitoring

```
title: SENTINEL APEX - CDB-ADVISORY Exploitation Attempt
id: cdb-cdb-advisory-det
status: experimental
description: Detects exploitation indicators related to CDB-ADVISORY.
references:
- https://intel.cyberdudebivash.com
author: CyberDudeBivash SENTINEL APEX
date: 2026/06/20
tags:
- attack.initial_access
- attack.t1190
logsource:
category: webserver
detection:
keywords:
- 'CDB-ADVISORY'
condition: keywords
falsepositives:
- Security testing / penetration testing
level: medium
```

Microsoft Sentinel - KQL Query

```
// SENTINEL APEX - SurrealDB: Indexed ORDER BY leaks the value ordering of a SELECT-restricted field
// Advisory | Microsoft Sentinel KQL
let time_window = ago(24h);
SecurityAlert
| where TimeGenerated > time_window
| where Description has "SurrealDB: Indexed ORDER BY leaks the va"
or AlertName has "SurrealDB: Indexed ORDER BY leaks the va"
or ExtendedProperties has "SurrealDB: Indexed ORDER BY leaks the va"
| extend Rule = "APEX-ADVISORY", Severity = "MEDIUM"
| project TimeGenerated, AlertName, AlertSeverity, Entities, Description, Rule, Severity
| order by TimeGenerated desc
```

Incident Response Playbook

Phase 1 - 0-24 Hours (Containment)

1. Activate IR team; assign lead analyst and executive sponsor.
2. Isolate or WAF-shield SurrealDB: Indexed ORDER BY leaks the value ordering of a SELECT instances exposed to the internet.
3. Enable verbose access logging on all SurrealDB: Indexed ORDER BY leaks the value ordering of a SELECT endpoints immediately.

- 4. Pull last 72h of web server and application logs for forensic baselining.
- 5. Check SENTINEL APEX IOC feeds and SIEM alerts for exploitation hits in your environment.
- 6. Notify relevant internal stakeholders and legal / compliance team.

Phase 2 - 24-72 Hours (Eradication)

- 1. Apply vendor patch for SurrealDB: Indexed ORDER BY leaks the value ordering of a SELECT or implement recommended mitigation.
- 2. Deploy Sigma and KQL detection rules from Section 6 across all SIEM platforms.
- 3. Rotate all API keys, service account credentials, and session tokens.
- 4. Conduct forensic timeline reconstruction for potential compromise window.
- 5. Validate patch deployment across 100% of affected instances via asset inventory.

Phase 3 - 7 Days (Recovery & Hardening)

- 1. Conduct post-incident review and update incident response runbooks.
- 2. Threat hunt for lateral movement or persistence artifacts using MITRE ATT&CK mapping.
- 3. Update asset inventory with patched version information and closure evidence.
- 4. Submit findings to internal risk register and CISO dashboard.
- 5. Schedule follow-up vulnerability scan to confirm remediation completeness.

Financial Impact Analysis (FAIR Model)

\$150K - \$900K Breach Cost Range (FAIR)	\$380K IBM Cost of Breach 2025 Median	\$120K/day Business Interruption Cost	\$85K Regulatory Fine Exposure
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Medium severity events require internal investigation and targeted customer notification. Average cost includes 340 hours of engineering time.

Remediation Cost Estimate: \$55K · Includes engineering time, IR retainer activation, forensic investigation, and customer notification costs.

Regulatory & Compliance Implications

Organizations processing this advisory must evaluate obligations under applicable regulatory frameworks. The table below maps this threat to relevant compliance requirements.

Framework	Reference	Obligation
PCI-DSS v4.0	Req. 6.3.3	Non-critical patches applied within 3 months
NIST CSF 2.0	ID.RA / PR.PS	Vulnerability assessment and platform security policies
ISO 27001:2022	A.8.8	Vulnerability management programme with documented remediation timelines

Actionable Recommendations

Prioritized defensive actions based on this period's intelligence.

1. Monitor SurrealDB: Indexed ORDER BY leaks the value ordering of a SELECT-restricted field for escalation.
2. Apply patches in next maintenance window.
3. Apply vendor patch for SurrealDB: Indexed ORDER BY leaks the value ordering of a SELECT-restricted field immediately - check NVD for official fix guidance.
4. Enable enhanced logging and alerting on all systems running affected software versions.
5. Deploy the Sigma detection rule from this report into your SIEM platform (Sentinel / Splunk / Elastic).
6. Audit all internet-facing instances of affected products and confirm patch deployment.
7. Conduct a threat hunt using MITRE ATT&CK techniques mapped in this advisory.

Appendix - Report Metadata & Legal

Field	Value
Report ID	intel--db55dbb8925c24225b1828a7
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Platform	CYBERDUDEBIVASH SENTINEL APEX v166.0 GOD-MODE
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Customer Email	-
TLP	TLP: CLEAR
STIX Version	2.1
ATT&CK Version	v15

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